

Yuantao Deng

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EDUCATION

New York University, NY/SH	Sep 2026-Present
Ph.D. in Neural Science, Advisor: Sukbin Lim	
Xi'an Jiaotong University, Xi'an	Sep 2022-Jun 2026
B.S. in Mathematics and Applied Mathematics	
Cornell University, NY	Jan-May 2025
Exchange Student	
University of California, Berkeley, CA	Jun-Aug 2024
Visiting Student	

Research Interests: Applied Mathematics, Computational Neuroscience

RESEARCH EXPERIENCE

Serial Dependence in Visual Working Memory	Jun-Aug 2026
Graduate Summer Research	Advisor: Sukbin Lim

- Implemented a computational model combining history-dependent efficient coding with attractor-based memory to investigate attractive and repulsive serial biases in visual working memory.
- Qualitatively reproduced 1-back attraction and 2–6-back repulsion under short-delay conditions, and identified delay-dependent discrepancies motivating multiscale history kernels and explicit modeling of memory uncertainty.

Multiscale Analysis and Dynamical Modeling of Rat fUS Hemodynamics	Jan-Jun 2026
Undergraduate Thesis	Advisor: Junjie Jiang and Zhiqiang Wei

- Processed a public rat functional ultrasound dataset using template construction, diffeomorphic registration, SIGMA atlas-based ROI mapping, and extraction of CBV-related time series.
- Characterized hemodynamic changes using PSD, fALFF, static and dynamic functional connectivity, Hurst exponents, and PCA; identified enhanced low-frequency activity and reorganized network dynamics 24–48 h after LPS treatment.
- Developed a delayed coupled-oscillator model with a vascular observation layer and applied approximate Bayesian computation for parameter inference; 91% of all empirical features fell within the simulated 1st–99th percentile ranges.

Large-Scale Functional Coupling and Network Dynamics during Working Memory	Apr 2025-May 2026
Westlake University	Advisor: Yuchen Xiao

- Analyzed intracranial local field potential recordings from 1,652 sites across 36 patients to characterize large-scale network dynamics during working-memory recognition, encoding, and retrieval.
- Quantified condition-dependent phase-amplitude and phase-phase coupling and frequency-specific information flow using directed phase transfer entropy, permutation testing, and multiple-comparisons correction.
- Applied time-delay-embedded hidden Markov models (TDE-HMMs) to identify task-locked latent network states, compare condition-dependent state-occupancy trajectories, and quantify state-resolved, frequency-specific long-range coherence.
- Built a modular Python pipeline for event alignment, artifact rejection, quality control, and figure generation; contributed to data analysis and manuscript preparation for the resulting bioRxiv preprint.

**Undergraduate Research Assistant at Weinan Sun Lab
Cornell University**

Jan-May 2025

- Evaluated and integrated image-registration and preprocessing algorithms for two-photon mesoscope data, benchmarking performance using signal-to-noise ratio, frame-to-frame stability, and downstream decoding accuracy to support a standardized, reproducible analysis pipeline.
- Implemented and evaluated large language model (LLM) methods, producing reproducible demos and technical notes to support method selection and subsequent analyses.

SoftHebb-Based Monocular Depth Estimation

Jun-Oct 2024

- Designed a hybrid ResNet–U-Net architecture incorporating a SoftHebb layer for monocular depth estimation from RGB images.
- Explored Hebbian co-activation-based updates for adapting intermediate feature representations within the depth-estimation pipeline.
- Resulted in a publication in *Applied and Computational Engineering*, 82(1), 65–70 (2024); see Publications.

Neural Networks for Solving Ordinary Differential Equations

Mar-May 2024

- Implemented Physics-Informed Neural Networks (PINNs) using automatic differentiation to solve ordinary differential equations by minimizing governing-equation residuals subject to initial and boundary conditions.
- Benchmarked PINN solutions against SciPy numerical solvers using relative L_2 error, residual norms, and initial- and boundary-condition violations.

PUBLICATIONS

- Zhang, W.*, Zhang, H.*, **Deng, Y.**, et al. (2026). Large-scale functional coupling and computational modelling reveal frontotemporal hotspots in distributed network connectivity during working memory recognition, encoding, and retrieval. *bioRxiv* [Preprint]. <https://doi.org/10.64898/2026.07.23.740204>
- **Deng, Y.** (2024). Adaptive depth estimation via SoftHebbLayer in a hybrid ResNet–UNet architecture. *Applied and Computational Engineering*, 82(1), 65–70. <https://doi.org/10.54254/2755-2721/82/20241264>

PERSONAL SKILLS

Languages: Mandarin (native), English (fluent)

Programming Skills: Python, C, C++, MATLAB